GARDE CITY
UNIVERSITY

NAME : Haobijam Tejkumar singh

ROLL NUMBER : 25BTCE206

BRANCH BCS

COURSE NAME : CMAT

HANDS ON : 3

DATE OF ISSUE : 04/12/2025

DATE OF SUBMISSION :24/2/2026

SUBJECT : COMPUTATIONAL MATHEMATICS

Handson session 3

1 .if a =x2Y+yz2thenfindV 20

Ans•. V 2∅=∂ 2∅∂ x2+ô 20/0 y2+ô 20/0 a

∂ 20/0 2xy = 2y

∂ 20/0 2yz=2y

V 20=2v+2v=4

2.if a =sinx+cosy+z2 then findV 20

Ans:V 2∅=∂ 2∅∂ x2+ô 20/0 y2+ô 20/0 a

∂ 20/0 -sinx

∂ 20/0 -cosy

∂ 20/0 2

V 20 =-sinx +(-cosy)+ 2

3.if a =ex+y2+logz thenfindV 20

Ans: v 20=0 x2+ô 20/ô y2+ô 20/ô a

∂ 20/0 x2=ex

∂ 20/ô y2 =2y = 2

∂ 20/0 z2=1z =-Iz2

v 20 = 0<+2-1 a

4. if =x3-3xy2 then show that is harmonic

Ans•. V 20=ô 20ô x2+ô 20/ô y2+ô 20/ô a

∂ 20/0 x2=6x ∂ 20/ô y2=-6x

∂ 20/0 z2=o v 20 , a isharmonic

5.if a =ex.cosythenfindV 20

Ans•. V 20=ô 20ô x2+ô 20/ô y2+ô 20/ô a

∂ 20/ô x2=ex ∂ 20/ô y2=cosy

∂ 20/ô z2=o

6.if a =x2-y2 then find V 20

Ans•.V 20=ô 20ô x2+ô 20/ô y2+ô 20/ô a

∂ 20/ô x2=2∂

20/ô y2=-2

∂ 20/ô z2=o

7.find Laplacian of r = x2+y2+z2

Ans•. V 2r=ô 2rô x2+ô 2r/ô y2+ô 2r/ô a

ô rô x=1/2x2+y2+z2 = 2x = xr

...applying d(u/v) = v.du - u.dv/vA2

∂ 20/0 r.l -x.ô rô x/rA2= r -x. w / =

= r2-x2r/r2 = x2+y2/r3

∂ 2r/ô x2=x2+y2/r3 ∂

2r/ô x2+y2/r3

∂ 2r/ô z2=x2+y2/r3

V 2r=ô 2rô x2+ô 2r/ô y2+ô 2r/ôz2=

= 2(x2+y2+z2)/r3

= 2r2/r3

= 2r

8. Q)Find V if
ϕ=ex2y2z2.

ANS:

∇ ϕ=∂ ϕ∂ x∂ ϕ∂ y∂ ϕ∂

∂ ϕ∂ x=2x ex2y2z2

∂ ϕ∂ y=2y ex2y2z2

∂ ϕ∂ z=2z ex2y2z2

∇ ϕ=2ex2y2z2xiyjzk

∇ ϕ=2ex2y2z2(xi+yj+zk) z

9. Q)Find div F and curl F if
F=xy2 i+2x2yz j-3yz2 k.

ANS•.

div F=∂ ∂ x(xy2)+∂ ∂ y(2x2yz)+∂ ∂ z(-3yz2)
div F=y2+2x2z-6yz div F=y2+2x2z-6yz curl
ijkö ö xö ö yö ö zxy22x2yz-3yz21
=i3z22x2y+k4xyz2xy curl F=(-3z2-
2x2y)i+(4xyz-2xy)k

10. Find div F and curl F if

$$F = x^3 i + y^3 j + z^3 k.$$

ANS:div F=∂ ∂ x(x3)+∂ ∂ y(y3)+∂ ∂ z(z3)
div F=3x2+3y2+3z2 divF=3(x2

curl ijkô yô zx3y3z31 =i∂z3ô ya y3ô z-p
z3ô x3ô z+kô y3ôx∂ x3

=i(0-0)-j (0-0)+k(0-

curlF=0

11. Find the directional derivative of

ϕ=4xz2+x2yz

at the point 1 ,-2,1.

ANS:

V xö yö z ∂ ϕ∂

x=4z2+2xyz ∂ ϕ∂y=x2z

∂ ϕ∂z=8xz+x2y

V 4)=(4z2+2xyz)i+(x2z)j+(8xz+x2y)k

∇ ϕ(1,-2,1)=(4-4)i+1j+6k

j+6k

12. lf ϕ = x^2y+ yz^2, then find∇ ²ϕ

ANS•.V 2ф=д 2фё x2+ё 2фё y2+ё 2фё z2

д Фё д x(x2y)=2xy д
2фё x2=д дx(2xy)=2

д Фё ё y(x2y+z2y)=x2+z2 д
2фё y2=o

д Фё д z(yz2)=2yz д 2фё
z2=2y

∇ 2ф=2y+O+2y

∇ 2ф=4y